



r/linuxquestions • 2 yr. ago

Cephalon_Zeash



What's the difference between a daemon and a service?

I've done some googling but the answers are fairly vague. For instance, what's the difference between the Docker daemon and the Docker service? Same with ssh.



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59

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Otaehryn • 2y ago

On Linux a daemon or service are interchangeable, although with systemd you can have oneshot service that runs something and then terminates. A daemon is always a process running in the background.

60 ...



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-George--- • 2y ago

I see that some answers are getting fancy with explanations involving `systemd`.

Both words predate `systemd` by decades.

Or whether they are "always doing something" or "waiting to do something" - the dumbest thing I've heard so far today.

"Daemon" originated in the unix world, "Service" in Windows. Both refer to programs that run and stay running, usually in a way that the user doesn't interact with (eg in a different user context, often more privileged, eg root or SYSTEM).

If it's designed just run occasionally or on a schedule or trigger and then quit, then yes terminology gets fuzzy and gray. Someone said that's still a service/daemon. Maybe - but many IT pros, possibly most, would disagree - even though such programs may in fact be launched by Windows "Service Manager" or `systemd`. (The former is often grossly misused/abused. The latter has never been limited to dealing with just daemons.) But sure, there's room for debate, and agreeing on a hard distinction isn't going to make or break society.

There are countless ways to launch tasks at the system level that do something and then quit - eg Windows Task Scheduler, which is complex and heavily used. Countless things stuffed into Windows Service Manager should instead go in there (possibly because Task Scheduler is honestly complex, fiddly, and defined tasks are prone to breaking with system changes). Or with Linux, you have crontab, cron.daily (etc.), etc. *Nobody* - at least nobody that matters or that I'm aware of - considers those "services" or "daemons". More like "Tasks", "Jobs", "Cronjobs", etc.

⊖ ↑ 19 ↓ ...



edman007 • 2y ago

I would argue that a daemon is a process that runs in the background all the time to provide a service.

A service however need not be a daemon, you have the `systemd` one shot services, but also an `inetd` http service in which case `initd` starts the service only when a request comes in and then it exits after providing it's service.

↑ 4 ↓ ...

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humanplayer2 • 2y ago

Thank you for asking this question. I learned from the answers!

↑ 3 ↓ ...

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PCChipsM922U • 2y ago

service/init managers, then they just changed the name to services.

↑ 3 ↓ ...



BCMM · 2y ago · Edited 2y ago

They are mostly synonyms. Historically, the Windows world preferred "service" and the Unix world preferred "daemon".

However, now we have systemd, and systemd consistently uses "service", at least when talking about its own configuration. Sometimes, people might use "daemon" to describe the long-running process itself, and "service" to describe the systemd configuration that controls it.

↑ 3 ↓ ...



linuxhiker · 2y ago · Edited 2y ago

A daemon provides the service.

The service (usually) is controlled by systemd

⊖ ↑ 6 ↓ ...



Michaelmrose · 2y ago

A demon provides a pitchfork. A **daemon** is just another name for service.

↑ 4 ↓ ...



Atemu12 · 2y ago

In my view, a daemon is just a kind of process that forks and keeps running "in the background" (as a child process of the process which exec'd it) while the "foreground" process exits immediately allowing further execution of the parent program while the child keeps running "in the background".

A service encapsulates a process into a higher abstraction unit that has metadata, surrounding semantics (i.e. what should be done when it fails, what should be done before it starts and a whole lot more) and can be managed by a service manager like systemd.

A daemon can be part of a service and historically was but a service does not need to include a daemon nowadays.

A service's process(es) can keep running without technically forking into the background because systemd can deal with that. It's actually better to not daemonise because, if there are further processes spawned, it's clearer which one is the "main" process of a service. Dunno about other service managers.

↑ 2 ↓ ...



[deleted] · 2y ago

Fundamentalists enjoy services.



oz1sej · 2y ago

The answers are interesting, and surprise me. I've always learned that "a daemon is a process which listens to a port". But that is perhaps no longer the case...?

↑ 2 ↓ ...



u/Zambito1 · 2y ago

To me a daemon is exactly one process while a service may consist of multiple processes. All daemons are services but not all services are daemons (a service may be a suite of daemons).

Also a daemon is long lived (runs until some error crashes it or it is manually stopped) while a service may be long lived or may do some job and die (such as an automatic update service checking for updates and terminating after it's done; must be started by some other long lived service like cron or systemd).

⊖ ↑ 0 ↓ ...



Michaelmrose · 2y ago

Making up definitions doesn't help.

↑ 0 ↓ ...

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⊕ **[deleted]** · 2y ago

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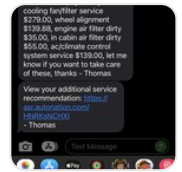
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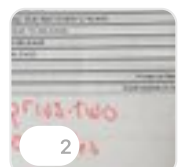
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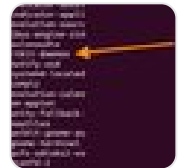
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